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Q-0000-001

Revision # 00

ITP step #: 6000-8

Unit : 4

Project: GMS

Equipment : Re-used part inspection and identification

The parts that were removed and to be removed consist of the coil brackets and corresponding hardware/accessories along with the rotor rim keys.

The rotor rim keys were already identified (number stamped mid-way on length of key)
The coil brackets were numbered using a piece of masking tape. The identification was completed according to rotor pole identification drawing (poles numbered clockwise, Pole #46 in line of upstream radial welded joint)
The rotor rim keys appeared in decent shape. We only had to wirewheel/sand some minor defects prior to reinstallation

The coil brackets also appeared in good condition. The coil bracket studs had a few dents/nicks that were repaired using a 1" dye-nut.

NCR #:

Measurement Instruments	T°		Measured by:	dd-mmm-yyyy
Visual	Before:	N/A	ANDRITZ Rep.: Jean-Luc Cote	17-Jun-2012
	After:	N/A	Cust. Repres.	Sept. 27, 2012

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ITP step #: 6000-9

Revision # 00

Project: GMS

Unit : 4

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Equipment : Rotor Pole Visual Inspection

The poles were in a bad condition at disassembly. The coils were partially covered in dust/carbon residues (likely brake dust). The insulation sheets between the coils appeared damaged

In some cases, the fiber insulating collar was "loose" and only retained in place by its fit around the pole body. The collars were only fixated by epoxy at two locations (top and bottom)

The pole bodies appeared in acceptable condition. They did not appear mechanically damaged.

NCR #:

Measurement Instruments	T°	Measured by:	dd-mmm-yyyy
Visual	Before: N/A		
	After: N/A	ANDRITZ Rep.: Jean-Luc Cote	30-May-2012
		Cust. Repres.	

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ITP step #: 6000-10

Revision # 00

Project: GMS

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Equipment : Rotor Pole Megger Testing at disassembly

For safety precautions, the megger testing was not completed in at disassembly due to the bad condition of the coils and the potential risk of asbestos

The poles were wrapped up prior to dismantling to ensure that asbestos encapsulated between the coils would not migrate around the powerhouse.

An electrical testing failure could have induce workers to friable asbestos exposure.

NCR #:

Measurement Instruments	T°	Measured by:	dd-mmm-yyyy
	Before: N/A		
	After: N/A	ANDRITZ Rep.: Jean-Luc Cote	30-May-2012
		Cust. Repres.	

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ITP step #: 6000-11

Project: GMS

Unit : 4

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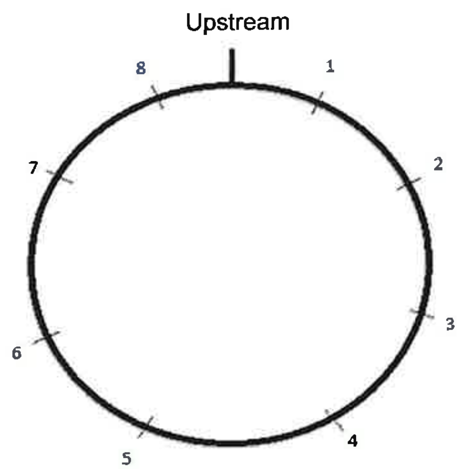
Readings in : original in mm

Rotor Levelling of unit 4 on service bay



#	Rdgs (mm)	Rdgs (inch)
1	0.07	0.003
2	0.09	0.004
3	0.06	0.002
4	0.04	0.002
5	0.08	0.003
6	0.00	0.000
7	0.05	0.002
8	0.05	0.002

***Note 1: Measured outside of coupling face**

***Note 2: The values were verified by Andritz using an optical level N3 with similar results**

***Note 3: The acceptance criteria on rotor leveling was 0.002" however due to extreme difficulty encountered by voith during levelling, Andritz engineering accepted 0.005"**

NCR #:

Measurement Instruments	T°		Measured by:	Andre Charbonneau (Voith)	dd-mmm-yyyy
Faro Laser Tracker	Before:		ANDRITZ Rep.:	C. Lemieux, J-L Cote	April 9th
X01000702349	After:		Cust. Repres.:		

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ITP step #: 6000-12

Revision # 00

Project: GMS

Unit : 4

Equipment : Rotor Pole concentricity, circularity and verticality

The as-found rotor pole concentricity, circularity, and verticality were not measured at disassembly. Due to the configuration, and measuring tools (lack of Faro Laser Tracker), we were not able to take the measurements as per ITP step 6000-12.

The information however (concentricity, circularity, and verticality) were previously obtained during the rotor sweep conducted in ITP step 6000-2

Concentricity: 0.004" on tolerance of 0.014"

Circularity: 0.019" on tolerance of 0.092"

Verticality: 0.026" on tolerance of 0.046"

NCR #:

Measurement Instruments	T°			dd-mmm-yyyy
	Before:	Measured by:		
	After:	ANDRITZ Rep.:	Jean-Luc Cote	30-Apr-2012
		Cust. Repres.		

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Q-0000-001

ITP step #: 6000-13

Project: GMS

Unit : 4

Revision # 00

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Equipment : RTD and air gap sensor cable as-found installation.

The RTD cables and upper air gap sensors were routed on top of the rotor fixated by a yellow epoxy (likely Loctite Hysol E120-HP). They were then routed down through the rotor rim key openings (bottom left picture)

The lower air gap sensor cables were routed up through the air duct between the rotor rim segments (top of right picture). The grounds were located at the far end of the rotor spider extremity. The cables inside the rotor spider were fixated by means of cable clamps submerged in silicone (bottom left picture). The cables then entered conduits fastened from the rotor spider rib arm to the slip ring shaft (bottom right picture).

NCR #:

Measurement Instruments	T°	Measured by:	dd-mmm-yyyy
	Before:		
	After:	ANDRITZ Rep.: Jean-Luc Cote	28-May-2012
		Cust. Repres.	

GMS G4 Rotor Pole As Found Measurements - May 28th 2012
ITP 6000-14 & 15
Measured by: Mike Rados
Instrument: Optical N3 #TL20435

Pole #	Dovetail A of old poles (mm)	Dovetail B of old poles (mm)	Average A&B (transferred in inch)	1/2 Dovetail Length (inch)	Distance between bottom of dovetail to BM (by N3 in inch)	Sum of E+F (inch)	Amount that each poles have to be raised to be even (inch)	Dovetail of new dovetails (mm)	Dovetail of new dovetails (inch)	1/2 new Dovetail Length (inch)	Sum of J+F (inch)	Amount that each poles have to be raised to be even (inch)
1	1981	1981	77.992	38.996	8.110	47.106	0.240	1981.2	78.000	39.000	47.110	0.200
2	1982	1982	78.031	39.016	7.875	46.891	0.455	1979.4	77.929	38.965	46.840	0.470
3	1980	1980	77.953	38.976	8.090	47.066	0.279	1980.4	77.969	38.984	47.074	0.236
4	1980	1980	77.953	38.976	8.020	46.996	0.349	1980.8	77.984	38.992	47.012	0.298
5	1980	1980	77.953	38.976	8.210	47.186	0.159	1980.4	77.969	38.984	47.194	0.116
6	1979	1979	77.913	38.957	8.090	47.047	0.299	1981.2	78.000	39.000	47.090	0.220
7	1982	1982	78.031	39.016	8.075	47.091	0.255	1980.4	77.969	38.984	47.059	0.251
8	1983	1983	78.071	39.035	8.050	47.085	0.260	1979.4	77.929	38.965	47.015	0.295
9	1978	1978	77.874	38.937	8.160	47.097	0.249	1980.3	77.965	38.982	47.142	0.168
10	1982	1982	78.031	39.016	8.270	47.286	0.060	1982.7	78.059	39.030	47.300	0.010
11	1982	1982	78.031	39.016	8.330	47.346	0.000	1979.4	77.929	38.965	47.295	0.015
12	1983	1983	78.071	39.035	8.145	47.180	0.165	1979.7	77.941	38.970	47.115	0.195
13	1980	1980	77.953	38.976	8.040	47.016	0.329	1980.2	77.961	38.980	47.020	0.290
14	1981	1981	77.992	38.996	7.850	46.846	0.500	1979.7	77.941	38.970	46.820	0.490
15	1981	1981	77.992	38.996	8.150	47.146	0.200	1979.6	77.937	38.969	47.119	0.191
16	1979	1980	77.933	38.967	8.300	47.267	0.079	1979.4	77.929	38.965	47.265	0.045
17	1984	1984	78.110	39.055	8.280	47.335	0.011	1979.7	77.941	38.970	47.250	0.060
18	1981	1981	77.992	38.996	8.310	47.306	0.040	1981.2	78.000	39.000	47.310	0.000
19	1978	1978	77.874	38.937	8.080	47.017	0.329	1979.4	77.929	38.965	47.045	0.265
20	1978	1978	77.874	38.937	8.210	47.147	0.199	1982	78.031	39.016	47.226	0.084
21	1984	1983	78.091	39.045	8.205	47.250	0.095	1979.7	77.941	38.970	47.175	0.135
22	1979	1979	77.913	38.957	8.130	47.087	0.259	1979.7	77.941	38.970	47.100	0.210
23	1983	1983	78.071	39.035	8.080	47.115	0.230	1980.4	77.969	38.984	47.064	0.246
24	1981	1981	77.992	38.996	8.155	47.151	0.195	1979.7	77.941	38.970	47.125	0.185
25	1982	1982	78.031	39.016	8.085	47.101	0.245	1979.4	77.929	38.965	47.050	0.260
26	1981	1981	77.992	38.996	8.120	47.116	0.230	1979.7	77.941	38.970	47.090	0.220
27	1979	1979	77.913	38.957	8.260	47.217	0.129	1981.8	78.024	39.012	47.272	0.038
28	1980	1980	77.953	38.976	8.190	47.166	0.179	1980.4	77.969	38.984	47.174	0.136
29	1979	1979	77.913	38.957	8.050	47.007	0.339	1981.4	78.008	39.004	47.054	0.256
30	1982	1982	78.031	39.016	7.990	47.006	0.340	1979.4	77.929	38.965	46.955	0.355
31	1981	1981	77.992	38.996	7.960	46.956	0.390	1980.2	77.961	38.980	46.940	0.370
32	1980	1980	77.953	38.976	8.070	47.046	0.299	1982.2	78.039	39.020	47.090	0.220
33	1979	1979	77.913	38.957	8.010	46.967	0.379	1980.4	77.969	38.984	46.994	0.316
34	1980	1980	77.953	38.976	7.965	46.941	0.404	1979.4	77.929	38.965	46.930	0.380
35	1979	1979	77.913	38.957	8.050	47.007	0.339	1979.4	77.929	38.965	47.015	0.295
36	1987	1987	78.228	39.114	8.030	47.144	0.202	1979.7	77.941	38.970	47.000	0.310
37	1983	1983	78.071	39.035	8.015	47.050	0.295	1981.2	78.000	39.000	47.015	0.295
38	1984	1985	78.130	39.065	8.205	47.270	0.076	1980.2	77.961	38.980	47.185	0.125
39	1981	1981	77.992	38.996	7.920	46.916	0.430	1981.2	78.000	39.000	46.920	0.390
40	1979	1979	77.913	38.957	7.990	46.947	0.399	1980.2	77.961	38.980	46.970	0.340
41	1977	1977	77.835	38.917	7.990	46.907	0.438	1979.7	77.941	38.970	46.960	0.350
42	1979	1979	77.913	38.957	7.935	46.892	0.454	1980.7	77.980	38.990	46.925	0.385
43	1981	1981	77.992	38.996	7.990	46.986	0.360	1981.2	78.000	39.000	46.990	0.320
44	1978	1978	77.874	38.937	7.900	46.837	0.509	1980.4	77.969	38.984	46.884	0.426
45	1983	1982.5	78.061	39.031	8.067	47.098	0.248	1979.4	77.929	38.965	47.032	0.278
46	1975	1976	77.776	38.888	8.000	46.888	0.458	1979.6	77.937	38.969	46.969	0.341
47	1979	1979	77.913	38.957	8.190	47.147	0.199	1979.4	77.929	38.965	47.155	0.155
48	1980	1980	77.953	38.976	8.120	47.096	0.249	1979.6	77.937	38.969	47.089	0.221

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ITP step #: 6000-16 & 17

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Equipment : Stator shape measurement

The stator shape is to be provided by BCH as per contract.

With regards to the above requirement, it shall be BC Hydro's responsibility to ensure the shape of the stator is within the tolerances set out in the CEA publication "Hydroelectric Turbine - Generator Units Guide for Erection Tolerances and Shaft System Alignment - Part V." Measurements of the stator shape shall be provided to the Contractor at a later date.

BCH also provided the required rotor magnetic center of the poles in relation to the coupling flange (26.302") in FAM-BCH-AH-081. The poles were adjusted accordingly at reassembly as documented in ITP step # 6300-15 & 16.

NCR #:

Measurement Instruments	T°	Measured by:		dd-mmm-yyyy
	Before:			
	After:	ANDRITZ Rep.:	Jean-Luc Cote	22-Aug-2012
		Cust. Repres.		

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Q-0000-001

ITP step #: 6000-18

Revision # 00

Project: GMS

Unit : 4

Equipment : Rotor Pole Key Height Dimensions As found

The rotor pole keys were above the rotor rim by 3.5". As a comparative, the rotor rim baffle plate hexagonal studs at 4" high. See picture below:

NCR #:

Measurement Instruments	T°	Measured by: Jean-Luc Cote	dd-mmm-yyyy
Before:			
After:		ANDRITZ Rep.: Jean-Luc Cote	20-Apr-2012
		Cust. Repres.	

ANDRITZ Hydro

GMS G4 - Rotor Rim Circularity and Concentricity June 21st
ITP 6000-19 & 6300-10
Measured by: Christian Maheux
Faro Laser Tracker X02000501577 (Calibrated March 29th 2012)

Summary of GMS G4 rotor rim measurements (in inch)

	Before Shrink	After Shrink	Tolerance
Circ Bottom	0.033	0.023	0.061
Circ Middle	0.025	0.027	0.061
Circ Top	0.033	0.026	0.061
Circ Total	0.023	0.024	0.046
Conc bottom	0.009	0.006	0.009
Conc Middle	0.010	0.005	0.009
Conc Top	0.011	0.003	0.009
Conc Total	0.010	0.005	0.009
Verticality	0.043	0.041	0.046

GMS G4 - Rotor Rim Circularity and Concentricity After Shrink June 21st 2012
ITP 6300-10
Faro Laser Tracker X02000501577 (Calibrated March 29th 2012)
Measured by Christian Maheux

Values in mm

	Bottom Rim, lower	Bottom Rim, upper	avg bottom	Middle Rim, Lower	Middle rim, Upper	avg middle	Top Rim, Lower	Top Rim, upper	Avg top	Total average
1	5231.30	5231.14	5231.22	5231.07	5231.22	5231.15	5231.22	5232.02	5231.62	5231.29
2	5231.41	5231.27	5231.34	5231.02	5231.23	5231.13	5231.12	5231.97	5231.55	5231.31
3	5231.39	5231.29	5231.34	5231.15	5231.19	5231.17	5231.10	5232.01	5231.56	5231.33
4	5231.43	5231.33	5231.38	5231.28	5231.30	5231.29	5231.15	5231.80	5231.48	5231.37
5	5231.35	5231.24	5231.30	5231.35	5231.15	5231.25	5231.20	5231.45	5231.33	5231.29
6	5230.85	5231.08	5230.97	5231.13	5231.16	5231.15	5231.03	5231.20	5231.12	5231.07
7	5230.79	5231.02	5230.91	5231.01	5231.15	5231.08	5230.97	5231.10	5231.04	5231.00
8	5231.02	5231.10	5231.06	5231.04	5231.02	5231.03	5231.03	5231.44	5231.24	5231.09
9	5231.10	5231.01	5231.06	5231.16	5231.25	5231.21	5231.12	5231.64	5231.38	5231.19
10	5231.24	5231.30	5231.27	5231.32	5231.23	5231.28	5231.19	5231.65	5231.42	5231.31
11	5231.14	5231.36	5231.25	5231.27	5231.22	5231.25	5231.13	5231.65	5231.39	5231.28
12	5231.08	5231.41	5231.25	5231.23	5231.20	5231.22	5231.06	5231.59	5231.33	5231.25
13	5230.67	5231.19	5230.93	5231.13	5231.11	5231.12	5230.82	5231.30	5231.06	5231.03
14	5230.73	5231.04	5230.89	5231.05	5231.16	5231.11	5230.92	5231.30	5231.11	5231.02
15	5231.04	5231.19	5231.12	5231.12	5231.23	5231.18	5231.05	5231.71	5231.38	5231.20
16	5230.97	5231.07	5231.02	5231.18	5231.24	5231.21	5231.20	5231.93	5231.57	5231.23
17	5231.07	5231.18	5231.13	5231.24	5231.34	5231.29	5231.25	5231.80	5231.53	5231.29
18	5230.91	5231.12	5231.02	5231.18	5231.27	5231.23	5231.23	5231.88	5231.56	5231.23
19	5231.02	5231.00	5231.01	5230.99	5231.20	5231.10	5231.04	5231.93	5231.49	5231.16
20	5230.82	5230.63	5230.73	5230.68	5230.98	5230.83	5230.75	5231.78	5231.27	5230.90
21	5230.84	5230.52	5230.68	5230.52	5230.88	5230.70	5230.71	5231.79	5231.25	5230.83
22	5231.16	5230.61	5230.89	5230.58	5230.85	5230.72	5230.87	5231.82	5231.35	5230.94
23	5230.92	5230.59	5230.76	5230.46	5230.93	5230.70	5230.63	5231.64	5231.14	5230.83
24	5230.85	5230.68	5230.77	5230.69	5231.02	5230.86	5230.84	5231.60	5231.22	5230.91
25	5230.97	5230.99	5230.98	5231.02	5231.19	5231.11	5231.01	5231.68	5231.35	5231.12
26	5230.68	5230.92	5230.80	5231.03	5231.11	5231.07	5231.02	5231.53	5231.28	5231.02
27	5230.62	5230.88	5230.75	5230.98	5231.08	5231.03	5230.97	5231.55	5231.26	5230.98
28	5230.76	5230.88	5230.82	5230.97	5231.08	5231.03	5230.95	5231.72	5231.34	5231.03
29	5230.90	5231.03	5230.97	5230.87	5230.99	5230.93	5230.91	5231.83	5231.37	5231.05
30	5230.56	5230.75	5230.66	5230.74	5230.84	5230.79	5230.76	5231.60	5231.18	5230.84
31	5230.68	5230.83	5230.76	5230.84	5230.85	5230.85	5230.76	5231.65	5231.21	5230.90
32	5230.97	5231.02	5231.00	5231.04	5230.96	5231.00	5231.07	5231.74	5231.41	5231.10
33	5230.95	5231.25	5231.10	5231.22	5231.08	5231.15	5231.13	5231.75	5231.44	5231.20
34	5231.01	5231.23	5231.12	5231.30	5231.22	5231.26	5231.14	5231.74	5231.44	5231.25
35	5230.81	5231.14	5230.98	5231.05	5231.11	5231.08	5230.96	5231.43	5231.20	5231.07
36	5231.01	5230.96	5230.99	5230.87	5230.91	5230.89	5230.77	5231.32	5231.05	5230.96
37	5230.53	5230.63	5230.58	5230.58	5230.67	5230.63	5230.63	5231.13	5230.88	5230.67
38	5230.64	5230.65	5230.65	5230.47	5230.71	5230.59	5230.57	5231.32	5230.95	5230.70
39	5231.14	5230.94	5231.04	5230.76	5230.86	5230.81	5230.67	5231.70	5231.19	5230.99
40	5231.22	5231.18	5231.20	5231.04	5230.92	5230.98	5230.86	5231.81	5231.34	5231.15
41	5231.08	5231.28	5231.18	5231.13	5231.16	5231.15	5231.00	5231.82	5231.41	5231.22
42	5230.88	5231.12	5231.00	5231.15	5231.16	5231.16	5231.01	5231.54	5231.28	5231.13
43	5230.94	5231.14	5231.04	5231.08	5231.14	5231.11	5231.02	5231.51	5231.27	5231.12
44	5230.58	5230.94	5230.76	5230.94	5230.89	5230.92	5230.88	5231.30	5231.09	5230.90
45	5230.62	5230.95	5230.79	5230.87	5230.88	5230.88	5230.86	5231.32	5231.09	5230.90
46	5231.00	5231.01	5231.01	5230.90	5230.99	5230.95	5231.00	5231.56	5231.28	5231.05
47	5230.81	5230.97	5230.89	5230.89	5230.94	5230.92	5230.87	5231.52	5231.20	5230.98
48	5231.02	5231.03	5231.03	5230.85	5230.96	5230.91	5231.01	5231.78	5231.40	5231.07

GMS G4 - Rotor Rim Circularity and Concentricity Prior to Shrink June 8th 2012
ITP 6009-19
Faro Laser Tracker X02000501577 (Calibrated March 29th 2012)
Measured by Christian Maheux

Values in mm

	Bottom Rim, lower	Bottom Rim, upper	avg bottom	Middle Rim, Lower	Middle rim, Upper	avg middle	Top Rim, Lower	Top Rim, upper	Avg top	Total average
1	5229.90	5229.95	5229.92	5229.88	5230.17	5230.02	5230.07	5230.56	5230.31	5230.06
2	5229.91	5229.88	5229.90	5229.87	5230.16	5230.02	5229.94	5230.39	5230.16	5230.01
3	5229.85	5230.08	5229.97	5229.87	5230.07	5229.97	5229.87	5230.29	5230.08	5230.00
4	5230.11	5230.09	5230.10	5230.09	5230.08	5230.08	5229.90	5230.31	5230.10	5230.09
5	5230.23	5230.19	5230.21	5230.20	5230.11	5230.16	5230.06	5230.21	5230.14	5230.17
6	5229.94	5230.19	5230.07	5230.17	5230.15	5230.16	5229.92	5230.07	5229.99	5230.08
7	5229.93	5230.16	5230.05	5230.05	5230.12	5230.09	5229.86	5229.95	5229.91	5230.03
8	5230.08	5230.18	5230.13	5230.10	5230.22	5230.16	5229.97	5230.15	5230.06	5230.12
9	5230.24	5230.27	5230.26	5230.33	5230.43	5230.38	5230.19	5230.31	5230.25	5230.30
10	5230.59	5230.73	5230.66	5230.66	5230.61	5230.63	5230.46	5230.57	5230.51	5230.61
11	5230.62	5230.93	5230.77	5230.79	5230.73	5230.76	5230.56	5230.63	5230.60	5230.73
12	5229.92	5230.49	5230.21	5230.60	5230.63	5230.61	5230.52	5230.71	5230.61	5230.46
13	5229.81	5230.57	5230.19	5230.60	5230.67	5230.64	5230.58	5230.81	5230.69	5230.49
14	5229.94	5230.51	5230.23	5230.64	5230.79	5230.72	5230.60	5230.88	5230.74	5230.54
15	5230.19	5230.65	5230.42	5230.66	5230.87	5230.76	5230.71	5231.08	5230.90	5230.67
16	5230.18	5230.71	5230.44	5230.74	5230.84	5230.79	5230.78	5231.17	5230.98	5230.71
17	5230.13	5230.56	5230.35	5230.66	5230.85	5230.76	5230.83	5231.10	5230.96	5230.65
18	5230.15	5230.60	5230.37	5230.65	5230.74	5230.69	5230.71	5231.08	5230.90	5230.62
19	5230.28	5230.51	5230.40	5230.51	5230.75	5230.63	5230.67	5231.05	5230.86	5230.60
20	5230.18	5230.15	5230.16	5230.28	5230.65	5230.47	5230.48	5231.05	5230.77	5230.43
21	5230.31	5230.07	5230.19	5230.17	5230.59	5230.38	5230.45	5231.00	5230.73	5230.40
22	5230.41	5230.20	5230.30	5230.25	5230.54	5230.39	5230.54	5231.02	5230.78	5230.46
23	5230.39	5230.36	5230.38	5230.35	5230.67	5230.51	5230.51	5230.85	5230.68	5230.50
24	5230.48	5230.61	5230.54	5230.60	5230.94	5230.77	5230.68	5231.02	5230.85	5230.70
25	5230.37	5230.78	5230.58	5230.76	5230.88	5230.82	5230.75	5230.96	5230.85	5230.74
26	5230.28	5230.72	5230.50	5230.81	5230.83	5230.82	5230.73	5230.97	5230.85	5230.71
27	5230.32	5230.76	5230.54	5230.72	5230.85	5230.79	5230.58	5230.93	5230.75	5230.69
28	5230.45	5230.70	5230.57	5230.73	5230.74	5230.73	5230.61	5230.96	5230.78	5230.69
29	5230.48	5230.75	5230.61	5230.63	5230.72	5230.68	5230.54	5230.94	5230.74	5230.67
30	5230.22	5230.54	5230.38	5230.51	5230.56	5230.54	5230.39	5230.80	5230.59	5230.49
31	5230.20	5230.55	5230.38	5230.43	5230.47	5230.45	5230.40	5230.75	5230.57	5230.45
32	5230.31	5230.54	5230.42	5230.51	5230.48	5230.49	5230.47	5230.69	5230.58	5230.49
33	5230.38	5230.65	5230.52	5230.61	5230.54	5230.58	5230.55	5230.66	5230.61	5230.56
34	5230.32	5230.71	5230.52	5230.65	5230.56	5230.60	5230.53	5230.67	5230.60	5230.57
35	5230.10	5230.61	5230.35	5230.47	5230.51	5230.49	5230.41	5230.42	5230.42	5230.42
36	5229.76	5230.41	5230.09	5230.31	5230.38	5230.34	5230.29	5230.38	5230.34	5230.25
37	5229.70	5230.15	5229.93	5230.13	5230.29	5230.21	5230.14	5230.33	5230.24	5230.11
38	5229.82	5230.09	5229.96	5230.02	5230.07	5230.05	5230.02	5230.48	5230.25	5230.06
39	5230.09	5230.13	5230.11	5230.01	5230.24	5230.13	5230.00	5230.53	5230.26	5230.15
40	5230.05	5230.10	5230.08	5230.09	5230.25	5230.17	5230.02	5230.52	5230.27	5230.16
41	5230.07	5230.30	5230.18	5230.32	5230.48	5230.40	5230.38	5230.80	5230.59	5230.36
42	5230.04	5230.48	5230.26	5230.46	5230.63	5230.55	5230.46	5230.82	5230.64	5230.46
43	5230.04	5230.50	5230.27	5230.48	5230.55	5230.51	5230.52	5230.78	5230.65	5230.46
44	5229.77	5230.32	5230.05	5230.43	5230.57	5230.50	5230.50	5230.79	5230.65	5230.37
45	5229.70	5230.32	5230.01	5230.34	5230.59	5230.46	5230.48	5230.79	5230.63	5230.34
46	5229.88	5230.21	5230.05	5230.19	5230.57	5230.38	5230.41	5230.77	5230.59	5230.31
47	5229.78	5230.00	5229.89	5229.95	5230.23	5230.09	5230.22	5230.70	5230.46	5230.11
48	5229.77	5229.88	5229.82	5229.87	5230.13	5230.00	5230.16	5230.74	5230.45	5230.04

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6000-20

Revision # 00

Project: GMS

Unit : 4

Page 1 of 1

Equipment : Rotor Pole Key Height Dimensions As found

The existing rotor pole driven keys were 92-13/16" length (average)
The existing rotor pole stationary keys were 88-13/16" length (average)
The taper was measured at 0.005"/inch
There was a 0.060" shim inserted in each dovetail, likely to compensate for
design miscalculations on the total thickness (driven + stationary keys)

NCR #:

Measurement Instruments	T°	Measured by:		dd-mmm-yyyy
	Before:			
	After:	ANDRITZ Rep.:	Jean-Luc Cote	4-Jun-2012
		Cust. Repres.		

ANDRITZ Hydro

Installation and Commissioning

Quality Control Record

ITP closure

Q-0000-002

ITP step #: 6000-21

Revision # 00

Project: GMS

Unit : 4

THIS FORM MUST BE COMPLETED BY THE QA ADVISOR OR HIS REPRESENTATIVE.

THIS FORM MUST BE COMPLETED AT THE END OF THE WORK FOR EACH SECTION OF THE ITP.

ALL NON-CONFORMANCES, DEFICIENCIES, ETC. MUST BE CORRECTED PRIOR TO THE PREPARATION OF THIS REPORT.

THIS INSPECTION COVERS ONLY THE ITEMS SPECIFIED IN THE CONTRACT.

	NO	YES	N/A
ALL WORK IS COMPLETED AS PER ERECTION PROCEDURES AND DRAWINGS.			X
THE FINAL PRODUCT HAS BEEN INSPECTED AT THE END OF THE JOB AND CONFORMS WITH DRAWINGS AND SPECIFICATIONS.			X
ANY NON-CONFORMANCES HAVE BEEN CORRECTED AS PER THE APPROVED DISPOSITIONS (write the NC's numbers below this table).			
ALL QA REPORTS ARE COMPLETED AS PER THE ITP.		X	
ALL FORMS HAVE BEEN REVISED AND APPROVED BY THE QA REPRESENTATIVE.		X	
ENGINEERING APPROVAL HAS BEEN OBTAINED WHENEVER REQUIRED.		X	

Comments: SNC-GMS-04-001

SCM-GMS-04-001

NCR #:

Measurement Instruments	T°	Measured by:	dd-mmm-yyyy
	Before:		
	After:	ANDRITZ Rep.: Jean-Luc Cote	5-Jun-2012
		Cust. Repres.	